Clinical Decision Support Tools for Predicting Outcomes in Patients Undergoing Total Knee Arthroplasty: A Systematic Review.

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CLINICAL & AGENTIC AI TAKEAWAY

Clinical decision support tools show promise in predicting total knee arthroplasty outcomes, but more research is needed to confirm their clinical applicability and interpretation.

ABSTRACT

--- REPOSITORY & PUBLISHER WEB METADATA --- Affiliations: School of Engineering, The University of Newcastle, Callaghan, New South Wales, Australia; Centre for Rehab Innovations, The University of Newcastle, Callaghan, New South Wales, Australia; Hunter Medical Research Institute, New Lambton Heights, New South Wales, Australia. Keywords: Predictive model, Total Knee Replacement, Total Knee Arthroplasty, Patient-reported Outcomes, Functional Outcomes, Clinical Decision Tool Extended Abstract from Repository: Background Total knee arthroplasty is the standard surgical treatment for end-stage osteoarthritis. Although widely accepted as a successful procedure, approximately 30% of patients are not satisfied due to non-optimal postoperative outcomes. Clinical decision support tools that are able to accurately predict post-surgery outcomes would assist in providing individualized advice or services to help alleviate possible issues, resulting in significant benefits to both the healthcare system and individuals. Methods Five databases (Ovid Medline, Ovid EMBASE, CINAHL complete, Cochrane Library, and Scopus) were searched for the key phrases "knee replacement" or "knee arthroplasty" and "decision support tool," "decision tool," "predict∗ tool," "predict∗ model," "algorithm" or "nomogram." Searches were limited to peer-reviewed journal articles published between January 2000 and June 2019. Reference lists of included articles were examined. Authors came to a consensus on the final list of included articles. Results Eighteen articles were included for review. Most models reported low predictive success and inability to externally validate. Both candidate and final predictor variables were inconsistent between studies. Only 1 model was considered strongly predictive (AUROC >0.8), and only 2 studies were able to externally validate their developed model. In general, models that performed well used large patient numbers, were tested on similar demographics, and used either nonlinear input transformations or a completely nonlinear model. Conclusion Some models do show promise; however, there remains the question of whether the reported predictive success can continue to be replicated. Furthermore, clinical applicability and interpretation of predictive tools should be considered during development.

Keywords: Predictive model, Total Knee Replacement, Total Knee Arthroplasty, Patient-reported Outcomes, Functional Outcomes, Clinical Decision Tool

Study Overview & Background

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Methodology & Clinical Focus

Study Classification: systematic review. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

Clinical decision support tools show promise in predicting total knee arthroplasty outcomes, but more research is needed to confirm their clinical applicability and interpretation.

Extended Repository Abstract

Background Total knee arthroplasty is the standard surgical treatment for end-stage osteoarthritis. Although widely accepted as a successful procedure, approximately 30% of patients are not satisfied due to non-optimal postoperative outcomes. Clinical decision support tools that are able to accurately predict post-surgery outcomes would assist in providing individualized advice or services to help alleviate possible issues, resulting in significant benefits to both the healthcare system and individuals. Methods Five databases (Ovid Medline, Ovid EMBASE, CINAHL complete, Cochrane Library, and Scopus) were searched for the key phrases "knee replacement" or "knee arthroplasty" and "decision support tool," "decision tool," "predict∗ tool," "predict∗ model," "algorithm" or "nomogram." Searches were limited to peer-reviewed journal articles published between January 2000 and June 2019. Reference lists of included articles were examined. Authors came to a consensus on the final list of included articles. Results Eighteen articles were included for review. Most models reported low predictive success and inability to externally validate. Both candidate and final predictor variables were inconsistent between studies. Only 1 model was considered strongly predictive (AUROC >0.8), and only 2 studies were able to externally validate their developed model. In general, models that performed well used large patient

numbers, were tested on similar demographics, and used either nonlinear input transformations or a completely nonlinear model. Conclusion Some models do show promise; however, there remains the question of whether the reported predictive success can continue to be replicated. Furthermore, clinical applicability and interpretation of predictive tools should be considered during development.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

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